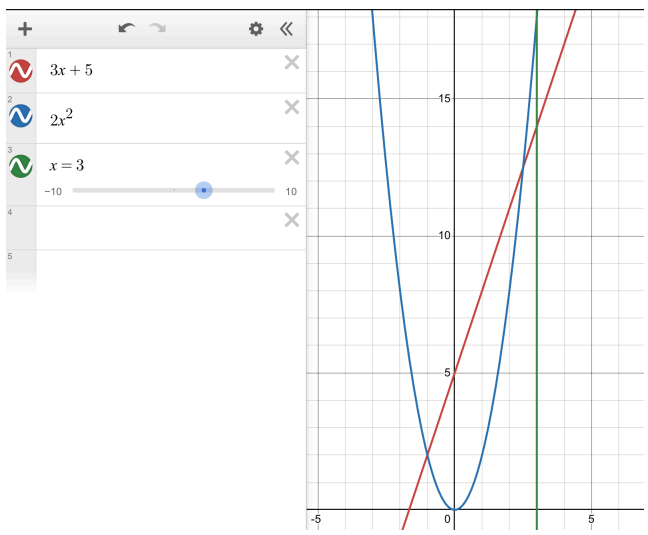
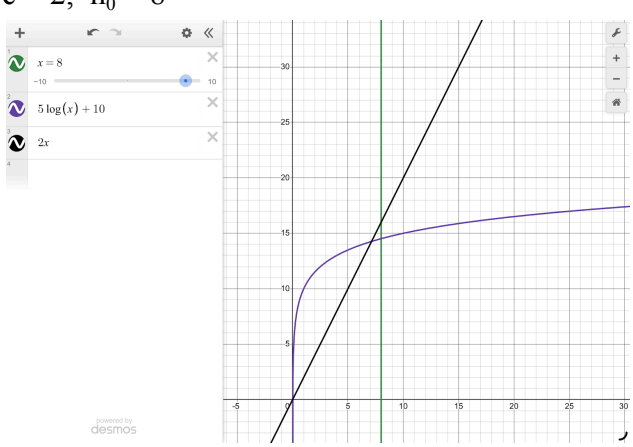
Gwynyn Hayes
Homework 3: Asymptotics
March 6th, 2025

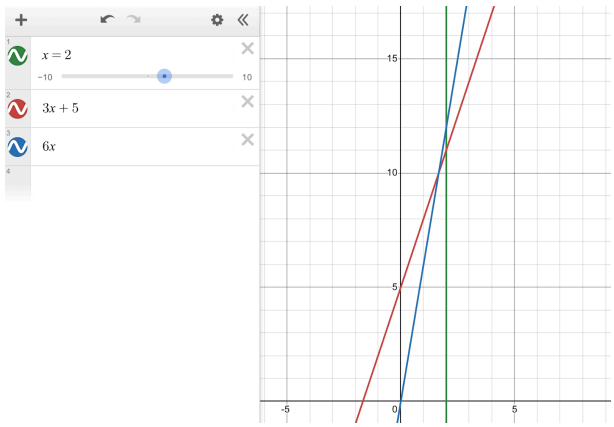
1. Complete the following definition:
   Let $T(n)$ be a function, and $f(n)$ be another function. We say that $T(n)$ is order $f(n)$, written O(f(n)), if there are constants $c < 0$ and $n_0 \geq 1$ such that for every integer $n \geq n_0$, $T(n) \leq c * f(n)$. When $T(n)=O(f(n))$, we say that $f(n)$ is an upper bound for $T(n)$.

2. For each pair of functions $f(n)$ and $g(n)$, use graphing to find positive numbers c and n_0 such, c that for every integer $n \geq n_0$, $f(n) \leq c \cdot g(n)$. Include your graphs in your submission; these graphs must effectively show that your choices for c and n_0 work.
   a. $f(n)=3n+5$, $g(n)=n^2$
      i. $c = 2$, $n_0 = 3$
      ii. 

   b. $f(n)=5\log(n)+10$, $g(n)=n$
      i. $c = 2$, $n_0 = 8$
      ii. 

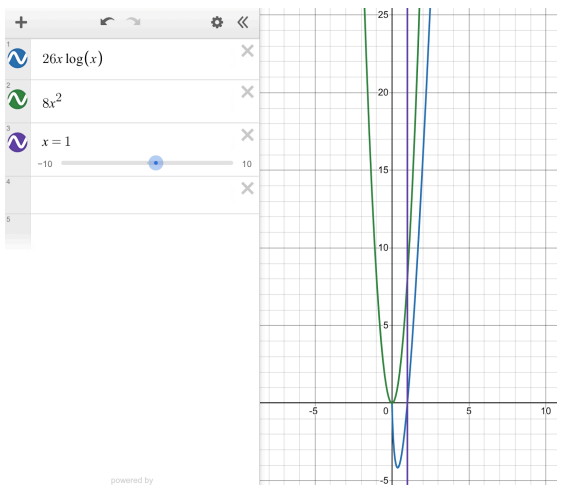
c. $f(n)=3n+5$, $g(n)=n$

i. $c = 6$, $n_0 = 2$

ii. 

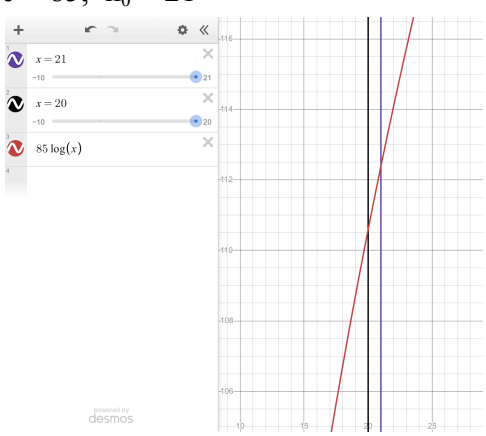
d. $f(n)=26n\log(n)$, $g(n)=n^2$

i. $c = 8$, $n_0 = 1$

ii. 

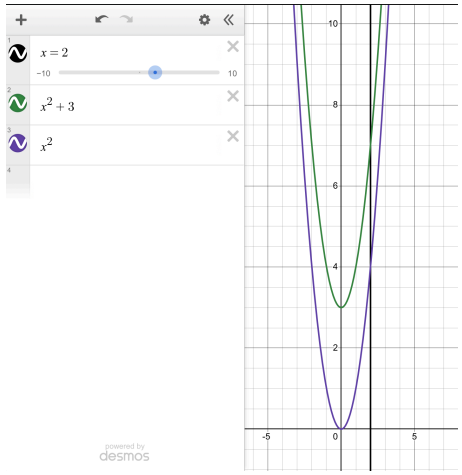
e. $f(n)=20$, $g(n)=\log(n)$

i. $c = 85$, $n_0 = 21$

ii. 

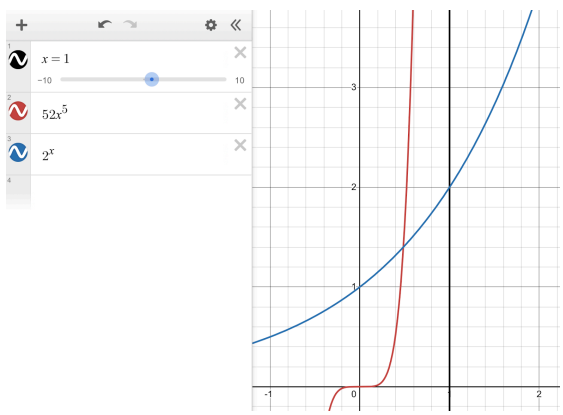
f. $f(n)=2n^2+3$, $g(n)=n^2$

i. $c = 1$, $n_0 = 2$

ii. 

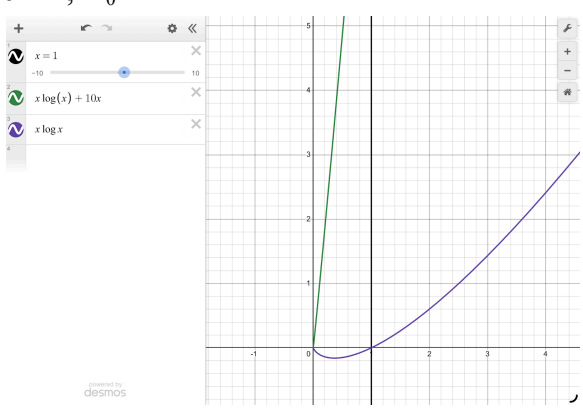
g. $f(n)=52n^5$, $g(n)=2^n$

i. $c = 1$, $n_0 = 1$

ii. 

h. $f(n)=n\log(n)+10n$, $g(n)=n\log(n)$

i. $c = 1$, $n_0 = 1$

ii. 

Questions 3-7 at end of PDF

3. Prove that n is order n^3.

4. Prove that $2n^2+4n+2$ is order n^4.
5. Prove that $3n^3-2n^2$ is order n^3.
6. Prove that n^3 is not order n^2.
7. Prove that n^2-n-1 is not order n.

8. (Puzzle Problem) Try to solve the following problem, and submit notes on your problem solving process (e.g., what ideas do you have? What did you try? What works? What doesn't work? etc.). You can earn full credit by demonstrating problem solving effort, even if you do not correctly solve the puzzle. You have a long piece of wood that you are cutting into n equal length pieces. With one cut, you can cut multiple pieces of wood (e.g., by stacking them on top of each other). What is the minimum number of cuts needed to get n equal size pieces? Write an algorithm for how you cut the wood to achieve this minimum.
    a. To get n equal size pieces we would cut each piece in half perfectly
        i. So to get 2 equal size pieces we would cut the log in half 1 time
        ii. 4 equal pieces would be 2 cuts
        iii. 8 pieces would be 3 cuts
        iv. For this pieces is doubling with each cut while cut is increasing by 1
    b. This only gets equal sized pieces that follow this pattern with each piece being cut each time a cut is made
    c. For an odd number of pieces each cut would make 2 initial cuts
        i. So to get 3 pieces we'd make 2 cuts
        ii. 6 pieces would add 3 cuts
    d. To get the minimum number of cuts its just the initial cuts, then for every additional number of pieces doubled there is one more cut.

Homework 3 : Asymptotics

Gwynnie Hayes

3. n is order n^3

let n_0 and c be positive integers such that $n \geq n_0$

then $n \leq c \cdot n^3$

let $n_0 = 1$ and $c = 1$

$n < n \cdot n \cdot n$ because $n \geq n_0 = 1$

$= n^3$

$= cn^3$ because $c = 1$

thus n is order n^2

4. $2n^2 + 4n + 2$ is order n^4

let $n_0 = 1$, $c = 6$

$2n^2 + 4n + 2 < 2n^2 + 4n$ because $n \geq n_0 = 1$

$\leq 2n^2 \cdot n \cdot n + 4n \cdot n \cdot n \cdot n$

$= 2n^4 + 4n^4$

$= 6n^4$

$= cn^4$ because $c = 6$

then $2n^2 + 4n + 2$ is order n^4

5. $3n^3 - 2n^2$ is order n^3

let $n_0 =$, $c = 3$

$3n^3 - 2n^2 < 3n^3$ because $n \geq n_0 = 1$

$= cn^3$ because $c = 3$

thus $3n^3 - 2n^2$ is order n^3

6. n^3 is not order n^2

$n^3 = n^2 \cdot n$

$\geq (c+1)n^2$ because $n \geq c+1$

$cn^2 + n^2$

$> cn^2$ because $n \geq c+1 > 1$ so $n^2 > 1$

So n^3 is not order n^2

7. $n^2 - n - 1$ is not order n

$n^2 - n - 1 > n^2 - n$

$> n^2$ because $n \geq 1$

$\geq n(c+1)$ because $n \geq c+1$

$= c \cdot n + n$

$> c \cdot n$ because $n \geq 1$

So $n^2 - n - 1$ is not order n